

QUIZ 4 (Dated: 31 Oct 2024)
Data Structures and Algorithm Analysis

Roll No: _____

Name: _____

Q1. Write a member function for class Doubly Header Circular Linked List to remove elements to make array in ascending order (You are not allowed to use any function)

```
void removeToSort(){
    for (DNode *t = head -> next ; t -> next != head ; )
        if (t -> val > t -> next -> val){
            DNode *toDel = t -> next;
            t -> next = t -> next -> next;
            delete toDel;
        }
        else t = t -> next;
}
```

Q2. For the Doubly header circular linear linked list, write member function to reverse the list by changing pointers only. (You cannot use stack)

```
void reverse(){
    if (head->next == head) return;
    DNode* cur = head, *temp = NULL;
    for ( ; !temp || cur != head ; cur = cur -> prev) {
        temp = cur -> next;
        cur -> next = cur -> prev;
        cur -> prev = temp;
    }
}
```

Q3. An integer vector has N persons want to receive the tickets. There are two class of tickets economy and business. In each turn, the counter man has either economy or business class ticket. The counter man has to find the first person in the list requires the current class of ticket. Write an efficient way to distribute the ticket. You may write pseudo code. Support, your answer, with computations that why your method is efficient.

```
declare two queues one for economy say eq and one for business class say bq
for each person in vector
    if person needs economy, push into eq otherwise in bq
while tickets not completed
    if ticket is economy
        select person from front of eq for ticket
        pop eq
    else
        select person from front of bq for ticket
        pop bq
```

The complexity is linear. In first linear loop, elements will be added in one of the two queues. In second linear loop (for tickets = persons), we will select person from one of the two queues.